PAU
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No
Date

Latihan Soal UAS

1.) V/div = 0,5 V
t/div = 0,5 ms → $0,5 \times 10^{-3}$
Sb. x = 4 kotak
Sb. y = 4 Kotak

Berapa Tegangan AC dan frekuensi?

=> V_{AC} = V/div × Sb. y
= 0,5 × 4
= 2 Vpp //

$f = \frac{1}{T} = \frac{1}{t/div \times Sb.X}$

$= \frac{1}{0,5 \times 10^{-3} \times 4} = \frac{1}{2 \times 10^{-3}} = \frac{1}{2} \times 10^{3} = 0,5 \times 10^{3}$
$= 500\ Hz$

2) a. $\varphi = \frac{a}{b} \times 360$
$= \frac{0,8}{4,9} \times 360 = 58,775°$ //

b. $Y_1 = 1,25 \quad Y_2 = 3$
$\varphi = \sin^{-1}\left(\frac{Y_1}{Y_2}\right)$
$= \sin^{-1}\left(\frac{1,25}{3}\right) = 24,624°$ //

3.) Jarak Gnd ke V_{DC} = 3 kotak
V_{DC} = V/div × Jarak Gnd ke V_{DC}
= 1 × 3
= 3 Vpp //

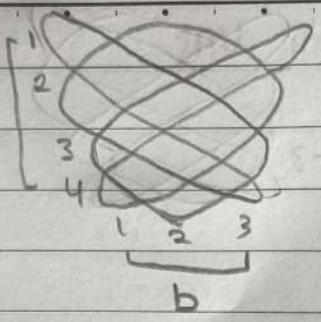
4.) Skala = 250, Batas Ukur (BU) = 15 mA, X = 120
$A = \frac{X}{Skala} \times BU = \frac{120}{250} \times 15\ mA = 7,2\ mA$ //

JOYKO® 30 Lines, 6 mm

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5.) a : b

4 : 3 //



Contoh :

400 Hz : 300 Hz

800 Hz : 600 Hz

1200 Hz : 900 Hz

1600 Hz : 1200 Hz

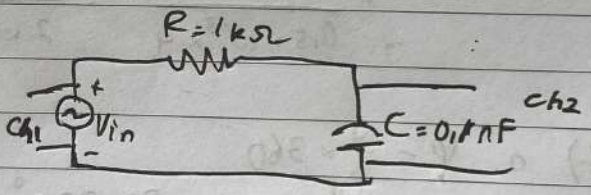
2 kHz : 1,5 kHz

6.) Dik : $V_{in} = 2 V_{PP}$

$R = 1000\ \Omega = 1 k\Omega$

$C = 0,1\ nF = 0,1 \times 10^{-9}$

$f = 1\ kHz$



Dit : φ ? V_{out} ?

Jwb : => $\omega = 2\pi f$

$= 2\pi \cdot 1000 = 6283,185$

$\therefore \varphi = -\tan^{-1} \omega RC$

$= -\tan^{-1}(6283,185 \times 1000 \times (0,1 \times 10^{-9}))$

$= -\tan^{-1}(6283,185 \times 10^{-7})$

$\varphi = -0,036°$

$\therefore V_{out} = \frac{1}{\sqrt{1^2 + (\omega RC)^2}}$

$= \frac{1}{\sqrt{1^2 + 6283,185^2 \times 1000^2 \times (0,1 \times 10^{-9})^2}}$

$= 1,999\ V_{PP}$ //